Muon does NOT bank Derived — reverts to IDENTIFIED/Insight after Cal §117 + Elie's blind-tau failure; tau stays FITTED; the Gindikin Γ_Ω is the promotion path (with a concrete π-signature test)

Keeper

2026-07-28

K966 — The muon does not bank; the honest resolution of the {24,71} pair

What came in

- Grace / Elie / Lyra folded the muon as DERIVED (pending Cal) — 24 = Γ(n_C), the domain's Gindikin gamma factor, "target-innocent in the sense you'd write Γ(n_C) before evaluating."
- Cal §117 (tire-kick): held the muon PROVISIONAL, not ratified. 24 is ≥4-fold expressible from the five integers (4!, 4·6, 6·4, 8·3), and even "Γ(5)" is 3-way mechanism-ambiguous (analytic Γ(5) vs $|S_4|$ vs |Weyl(SO(6))|, all = 24). So "24 = Γ(n_C)" is a real source ONLY if the muon mechanism produces a literal analytic Γ (zeta-reg / heat-kernel measure / volume), pinned independent of the value. The muon was ruled terminal-coincidence last week, so re-opening needs a NEW forced mechanism, not a Γ-costumed relabel. Cal's stated arbiter: because 24 is easy to hit and 71 is prime (hard), Elie's blind tau-forward decides both — 71 blind ⍰ mechanism real ⍰ muon+tau Derived; no 71 ⍰ muon's 24 was one of four ways ⍰ muon reverts.
- Elie (blind tau-forward): ran the vertex address on GF(128). 71 does NOT come out — not the primitive-element count, not the irreducible-polynomial count, not the field size. Its only route is "field dim + half the field" (g + 2^{g−1}), two pieces glued to hit a known number. Tau stays FITTED. (Elie: refused to dress the sum up as a mechanism — the fit-in-disguise trap.)
- Grace (form-ambiguity catch): the constants file carries the tau as $(24/\pi^2)^6$·(7/3)^{10/3}, NOT 49·71 — two competing forms, both with an un-sourced piece (71 decomposed; 10/3 un-sourced). Two forms crossing over ⍰ neither forced ⍰ Fitted regardless of which you pick (the glueball form-ambiguity pattern).

Ruling (both directions honest)

TAU → FITTED (confirmed). Elie's blind run failed to produce 71; the alternate form's 10/3 is equally un-sourced. Held at the floor. Promotion path = the stage-1 orbit→mass map (joint Lyra+Elie), producing 71 (or the 10/3) forward, blind.

MUON → IDENTIFIED / Insight (re-opened; NOT Derived). By Cal's own arbiter logic (tau-blind failed) AND by my pre-registered gate (2026-07-28: "banks only if Cal shows Γ(n_C) is the SOURCE, not merely that Γ(5)=24"), the muon does not bank as Derived. The mechanism producing a *literal analytic* Γ has not been exhibited; 24 is degenerate; the corroborating prime 71 failed; the terminal-coincidence prior sets the bar. Per [[feedback_clean_form_is_candidate_not_bank_until_mechanism]], a clean form with a plausible mechanism is a CANDIDATE, not a bank.

But — calibrate the OTHER way too (the muon is NOT dead-Fitted). The Gindikin Γ_Ω is a *real analytic object of* D_IV^5, not an invented costume, and there is a genuine, concrete, target-innocent promotion path (below). So the muon upgrades from K963's FITTED-lead to IDENTIFIED/Insight — the re-opening of the terminal-coincidence ruling IS justified (a real structural candidate appeared), even though the promotion to Derived is not. Elie was right to re-open; Cal is right it doesn't bank. Both honored.

Net count reverts: the team's ~17 Derived (with the muon) steps back to ~2 Proved + 16 Derived, muon as a strong Identified, tau + m_t-Value + the value-bearing angles as the honest tail. Grace/Elie/Lyra re-fold muon Derived→Identified (minor bookkeeping; the accuracy never moved).

★ THE PROMOTION PATH — a concrete, computable test (the research the team needs)

The Gindikin gamma of D_IV^5 (rank r=2, cone dim p=n_C=5, multiplicity d=p−2=3): > Γ_Ω(s) = (2π)^{3/2} · Γ(s) · Γ(s − 3/2).

Computed at the scalar address s = n_C = 5: > Γ_Ω(5) = (2π)^{3/2} · Γ(5) · Γ(7/2) = 45 · 2^{3/2} · π^2 (exact; 45 = g^2 − rank^2, the θ_{13} denominator).

The discriminator Cal demanded — the π-signature. $|S_4|$ = 24 and |Weyl(SO(6))| = 24 carry NO π. Analytic Γ_Ω carries the (2π)^{3/2} factor ⍰ π structure. The muon form is $(24/\pi^2)^6$ — it pairs the 24 *with* a π^2. A symmetry-count would never come with a π^2; analytic Γ_Ω does. So the very presence of π^2 alongside the 24 is (circumstantial) evidence the 24 is the analytic Γ(s) factor of Γ_Ω, not a discrete count — exactly the distinctive, un-fakeable signature the promotion needs.

★ Cal #27 / peak-convergence guard (fires HARDEST here): the 45 = g^2−rank^2 popping out and the clean π^2 are *seductive* and must NOT bank the muon. This is a LEAD, a promotion path — NOT a result. $24/\pi^2$ is $\Gamma(5)/\pi^2$, which is not Γ_Ω(5) (= 45·2^{3/2}π^2); the relation between the muon's ($24/\pi^2$) and the actual Γ_Ω must be derived at the muon's address, not noticed. The rigorous bar (Cal §117's four conditions) still gates it: 1. exhibit the muon derivation producing a literal analytic Γ_Ω (the π-signature entangling 24 with π^2 is the tell to make rigorous); 2. target-innocent address (why s and the 6th power — derived, not fit); 3. tau-71 (or 10/3) blind — currently FAILED; 4. a new forced mechanism re-opening the terminal-coincidence ruling.

The clincher test (unifies the sector): if Γ_Ω is the real lepton-mass mechanism, ONE Γ_Ω-based measure must produce e, μ, τ from three discrete-series addresses — and *predict* the tau (resolving the 71-vs-10/3 form-ambiguity Grace flagged). The tau form $(24/\pi^2)^6$·(7/3)^{10/3} = muon-form × boundary-correction is *consistent* with a shared Γ_Ω structure + address shift — so the lepton-unification computation resolves the muon Γ_Ω question AND the tau form-ambiguity at once.

Consequence — the whole flavor sector converges on ONE object

The muon-Γ_Ω address, the tau orbit→mass map, the quark-generation precision lead (K965/boundary-distance), the y_t deficit, and the generation count ALL reduce to the discrete-series / radial-address structure on D_IV^5. That is the convergence hub. Cal's reduction level is the linchpin — it unblocks Lyra's Gram-matrix radical (the generation count) and the address parametrization the flavor sector needs. Concentrate force there.

— K966, Keeper, 2026-07-28. Holds my own pre-registered bar; calibrates both directions (muon up from Fitted, not up to Derived; tau held). Γ_Ω(D_IV^5) computed exact. See [[Keeper_K963_BOTTOM_UP_TIER_REVIEW_apply_sharpened_
07-28]], [[Keeper_K964_CEILING_VALUE_DECOMPOSITION_of_the_Identified_tier_three_earning_conditions_2026-
07-28]], [[Keeper_K965_m_t_over_m_b_adjudication_HOLD_Identified_and_it_is_a_test_case_for_the_quark_generation_lead_
07-28]].